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Unit II:

Fuzzy Set & Fuzzy logic: (14M).

- Fuzzy Sets & Systems.
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- Relation between Crisp set & Fuzzy set.
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UNIT - II

FUZZY SET AND FUZZY LOGIC

INTRODUCTION

Fuzzy sets or fuzzy models are mathematical means of representing vagueness and imprecise information. These models have the capability of recongnising, representing, manipulating, interpreting and utilizing data and information that are vague and lack certainty.

Fuzzy logic is based on the observation that people make decisions based on imprecise and non-numerical information. Fuzzy logic is a form of many-valued logic in which the truth value of variables may be nay real number between 0 and 1. It is employed to handle the concept of partial truth where the truth value may range between completely true and completely false.

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Fuzzy set:

Let X be a non-zero set. A fuzzy set 'A' of X is defined as

$$A = \{\mu_A(x), x\}, x \in X$$

where $\mu_A : X \to [0,1]$ is a function called as the membership function of 'A' & $\mu_A(x)$ is the grade of membership or degree of belongingness or degree of membership of $x \in X$ in 'A'.

The fuzzy set 'A' can also be written as

$$A = \{(\mu_A(x_1), x_1), (\mu_A(x_2), x_2), \dots (\mu_A(x_n), x_n)\}$$

Symbolically, we can write

$$A = \left\{ \frac{\mu_A(x_1)}{x_1}, \frac{\mu_A(x_2)}{x_2}, \dots, \frac{\mu_A(x_n)}{x_n} \right\}$$

Crisp set:

Crisp set is often referred as a classical or ordinary set in the context of fuzzy set theory.

A set defined using a characteristic function that assigns a value of either 0 or 1 to each element of the universe, thereby discriminating between members & non-members is called a crisp set.

- The membership function can be used to define a set 'A' given by

$$\chi_A(x) = \begin{cases} 1 & , \text{ if } x \in A \\ 0 & , \text{ if } x \notin A \end{cases}$$

- The function χ (read as chi) is known as crisp membership function.
- The set of natural members, real numbers, integers, rational numbers are all examples of crisp set.

Operations & Combinations on fuzzy sets:

1) Equality of Two fuzzy sets:

Two fuzzy sets A and B are said to be equal written as $A = B$, iff $\mu_A(A) = \mu_B(x)$, which means that their membership values must be equal.

Let $X = \{a, b, c\}$.
Consider the fuzzy sets A, B, C of X defined by
$A = \{(0.3, a), (0.6, b), (0.7, c)\}$
$B = \{(0.6, b), (0.3, a), (0.7, c)\}$.
$C = \{(0.4, a), (0.5, b), (0.9, c)\}$.

Here $A = B$, but $A \neq C$, $B \neq C$

2) Complement or Absolute Complement of a Fuzzy set:

The complement or absolute complement of a fuzzy set 'A' is the set $\bar{A}$ with degree of the membership of an element $\bar{A}$ is equal to one minus the degree of membership of this element in A.

Let $A = \left\{ \frac{0.2}{3}, \frac{0.7}{4}, \frac{0.8}{6}, \frac{0.9}{7} \right\}$

$\bar{A} = 1 - \mu_A(x) = \left\{ \frac{0.8}{3}, \frac{0.3}{4}, \frac{0.2}{6}, \frac{0.1}{7} \right\}$.

3) Union of two fuzzy sets:

The union of two fuzzy sets A & B is the fuzzy set A∪B, where the degree of membership of an element in A∪B is the maximum of the degrees of membership of this element in A and in B.

Let $A = \left\{ \frac{0.3}{a}, \frac{0.9}{b}, \frac{0.6}{c}, \frac{0.4}{d} \right\}$ & $B = \left\{ \frac{0.4}{a}, \frac{0.8}{b}, \frac{0.2}{c}, \frac{0.7}{d} \right\}$

$A \cup B = \max \{ \mu_A(x), \mu_B(x) \}$

$= \left\{ \frac{0.4}{a}, \frac{0.9}{b}, \frac{0.6}{c}, \frac{0.7}{d} \right\}$.

4) Intersection of two fuzzy sets :

The intersection of two fuzzy sets A & B is the fuzzy set $A \cap B$, where the degree of membership of an element in $A \cap B$ is the minimum of the degrees of membership of this element in A & in B.

for eg: Let $A = \left\{\frac{0.3}{a}, \frac{0.9}{b}, \frac{0.6}{c}, \frac{0.4}{d}\right\}$ & $B = \left\{\frac{0.4}{a}, \frac{0.8}{b}, \frac{0.2}{c}, \frac{0.7}{d}\right\}$

$A \cap B = \min\{\mu_A(x), \mu_B(x)\}$

$= \left\{\frac{0.3}{a}, \frac{0.8}{b}, \frac{0.2}{c}, \frac{0.4}{d}\right\}$

5) Difference of two fuzzy sets :

The difference of two fuzzy sets A & B is defined as the intersection of first fuzzy set A and complement of second fuzzy set B. It is denoted by $\frac{A}{B}$ or $A - B$.

for eg.: Let $A = \left\{\frac{1}{1}, \frac{0.15}{1.5}, \frac{0.3}{2}, \frac{0.5}{2.5}, \frac{0}{3}\right\}$ & $B = \left\{\frac{1}{1}, \frac{0.6}{1.5}, \frac{0.2}{2}, \frac{0.1}{2.5}, \frac{0}{3}\right\}$

Here $\overline{B} = \left\{\frac{0}{1}, \frac{0.4}{1.5}, \frac{0.8}{2}, \frac{0.9}{2.5}, \frac{1}{3}\right\}$

$\therefore A - B = A \cap \overline{B} = \left\{\frac{0}{1}, \frac{0.15}{1.5}, \frac{0.3}{2}, \frac{0.5}{2.5}, \frac{0}{3}\right\}$

- The relative complement of A with B is B-A ie. $B \cap \overline{A}$
- The relative complement of B with A is A-B ie. $A \cap \overline{B}$

6) Algebraic sum of two fuzzy sets:
Let A & B be any two fuzzy sets, then their algebraic sum is denoted & defined by

$$\mu_{A+B}(x) = [\mu_A(x) + \mu_B(x)] - [\mu_A(x).\mu_B(x)]$$

for eg: Let $A = \left\{\frac{1}{1} + \frac{0.5}{1.5} + \frac{0.3}{2} + \frac{0.4}{2.5}\right\}$ & $B = \left\{\frac{0.4}{1} + \frac{0.2}{1.5} + \frac{0.7}{2} + \frac{0.1}{2.5}\right\}$

$\mu_{A+B}(1) = (1+0.4) - (1 \times 0.4)$
$= 1.4 - 0.4 = 1$

$\mu_{A+B}(1.5) = (0.5+0.2) - (0.5 \times 0.2) = 0.7 - 0.1 = 0.6$

$\mu_{A+B}(2) = (0.3+0.7) - (0.3 \times 0.7) = 1.0 - 0.21 = 0.79$

$\mu_{A+B}(2.5) = (0.4+0.1) - (0.4 \times 0.1) = 0.5 - 0.04 = 0.46$

$$A+B = \left\{\frac{1}{1} + \frac{0.6}{1.5} + \frac{0.79}{2} + \frac{0.46}{2.5}\right\}$$

7) Algebraic product of two fuzzy sets:
Let A & B be any two fuzzy sets. Then their algebraic product is denoted & defined by

$$\mu_{A\cdot B}(x) = \mu_A(x) \cdot \mu_B(x)$$

for eg. Let $A = \left\{\frac{1}{1} + \frac{0.5}{1.5} + \frac{0.3}{2} + \frac{0.4}{2.5}\right\}$ & $B = \left\{\frac{0.4}{1} + \frac{0.2}{1.5} + \frac{0.7}{2} + \frac{0.1}{2.5}\right\}$

$$A \cdot B = \left\{\frac{0.4}{1} + \frac{0.1}{1.5} + \frac{0.21}{2} + \frac{0.04}{2.5}\right\}$$

8) Bounded sum of two fuzzy sets:
The bounded sum of two fuzzy sets A and B is denoted & defined by

$$\mu_{A \oplus B} = \min\left[1, \mu_A(x) + \mu_B(x)\right]$$

for eg: Let $A = \left\{\frac{1}{1} + \frac{0.5}{1.5} + \frac{0.3}{2} + \frac{0.4}{2.5}\right\}$ & $B = \left\{\frac{0.4}{1} + \frac{0.2}{1.5} + \frac{0.7}{2} + \frac{0.1}{2.5}\right\}$

$\mu_{A \oplus B}(1) = \min[1, 0.4] = 0.4$

$\mu_{A \oplus B}(2.5) = \min[1, 0.7] = 0.7$

$\mu_{A \oplus B}(2) = \min[1, 1] = 1$

$\mu_{A \oplus B}(2.5) = \min[1, 0.5] = 0.5$

$\therefore A \oplus B = \left\{\frac{0.4}{1} + \frac{0.7}{1.5} + \frac{1}{2} + \frac{0.5}{2.5}\right\}$

9) Bounded Difference of Two fuzzy sets:

The bounded difference of two fuzzy sets A & B is denoted & defined by

$$\mu_{A \ominus B}(x) = \max[0, \mu_A(x) - \mu_B(x)]$$

Eg: Let $A = \left\{\frac{1}{1} + \frac{0.5}{1.5} + \frac{0.3}{2} + \frac{0.4}{2.5}\right\}$ & $B = \left\{\frac{0.4}{1} + \frac{0.2}{1.5} + \frac{0.7}{2} + \frac{0.1}{2.5}\right\}$

$\mu_{A \ominus B}(1) = \max[0, 0.6] = 0.6$

$\mu_{A \ominus B}(1.5) = \max[0, 0.3] = 0.3$

$\mu_{A \ominus B}(2) = \max[0, -0.4] = 0$

$\mu_{A \ominus B}(2.5) = \max[0, 0.3] = 0.3$

$$A \ominus B = \left\{\frac{0.6}{1} + \frac{0.3}{1.5} + \frac{0}{2} + \frac{0.3}{2.5}\right\}.$$

SOLVED EXAMPLES:

Ex.1. Let $X = \{a, b, c\}$. The fuzzy sets A and B of X are defined as

$$A = \left\{\frac{0.3}{a}, \frac{0.5}{b}, \frac{0.7}{c}\right\}, \qquad B = \left\{\frac{0.4}{a}, \frac{0.5}{b}, \frac{0.8}{c}\right\}.$$

Find: (i) $A \cup B$ (ii) $A \cap B$ (iii) $\overline{A}, \overline{B}$

(iv) Check whether $\overline{A} \cap A \neq \phi$ and $\overline{A} \cup A \neq X$ (v) $A - B$

Soln. Given: $X = \{a, b, c\}$,

$$A = \left\{\frac{0.3}{a}, \frac{0.5}{b}, \frac{0.7}{c}\right\},$$

$$B = \left\{\frac{0.4}{a}, \frac{0.5}{b}, \frac{0.8}{c}\right\}$$

(i) $A \cup B = \max\{\mu_A(x), \mu_B(x)\} = \left\{\frac{0.4}{a}, \frac{0.5}{b}, \frac{0.8}{c}\right\}$

(ii) $A \cap B = \min\{\mu_A(x), \mu_B(x)\} = \left\{\frac{0.3}{a}, \frac{0.5}{b}, \frac{0.7}{c}\right\}$

(iii) $\overline{A} = 1 - \mu_A(x) = \left\{\frac{0.7}{a}, \frac{0.5}{b}, \frac{0.3}{c}\right\}$

$$\overline{B} = 1 - \mu_B(x) = \left\{\frac{0.6}{a}, \frac{0.5}{b}, \frac{0.2}{c}\right\}$$

(iv) $\overline{A} \cap A = \min\{\mu_{\overline{A}}(x), \mu_A(x)\}$

$$= \left\{\frac{0.3}{a}, \frac{0.5}{b}, \frac{0.3}{c}\right\} \neq \phi$$

$$\overline{A} \cup A = \max\{\mu_{\overline{A}}(x), \mu_A(x)\}$$

$$= \left\{\frac{0.7}{a}, \frac{0.5}{b}, \frac{0.7}{c}\right\} \neq X$$

(v) $A - B = A \cap \overline{B} = \left\{\frac{0.3}{a}, \frac{0.5}{b}, \frac{0.3}{c}\right\}$ ....Ans.

Ex.2. **Consider the fuzzy sets A, B, C defined on the interval X = [0, 10] of real numbers by the membership grade function**

$$\mu_A(x) = \frac{x}{x+2}; \quad \mu_B(x) = 2^{-x}, \quad \mu_C(x) = \frac{1}{1+10(x-2)^2}.$$

Determine the mathematical formula for:

(i) $\overline{A}$ (ii) $A \cup B \cup C$

Soln.

x	$\mu_A(x) = \frac{x}{x+2}$	$\mu_B(x) = 2^{-x}$	$\mu_C(x) = \frac{1}{1+10(x-2)^2}$
0	0	1	0.024
1	0.33	0.5	0.09
2	0.5	0.25	1
3	0.6	0.125	0.09
4	0.66	0.0625	0.0243
5	0.71	0.0312	0.0109

6	0.75	0.0156	0.0062
7	0.77	0.0078	0.0039
8	0.8	0.0039	0.0027
9	0.818	0.0019	0.002
10	0.833	0.0009	0.0015

Here $A = \left\{\frac{0}{0}, \frac{0.33}{1}, \frac{0.5}{2}, \frac{0.6}{3}, \frac{0.66}{4}, \frac{0.71}{5}, \frac{0.75}{6}, \right.$

$\left. \frac{0.77}{7}, \frac{0.8}{8}, \frac{0.818}{9}, \frac{0.833}{10} \right\}$

$\therefore \overline{A} = \left\{\frac{1}{0}, \frac{0.67}{1}, \frac{0.5}{2}, \frac{0.4}{3}, \frac{0.34}{4}, \frac{0.29}{5}, \frac{0.25}{6}, \right.$

$\left. \frac{0.23}{7}, \frac{0.2}{8}, \frac{0.182}{9}, \frac{0.167}{10} \right\}$

& $A \cup B \cup C = \left\{\frac{1}{0}, \frac{0.5}{1}, \frac{1}{2}, \frac{0.6}{3}, \frac{0.66}{4}, \frac{0.71}{5}, \right.$

$\left. \frac{0.75}{6}, \frac{0.77}{7}, \frac{0.8}{8}, \frac{0.818}{9}, \frac{0.833}{10} \right\}$ **..Ans.**

Ex.4. Given that $A = \frac{0.8}{3} + \frac{0.9}{4} + \frac{0.6}{5}$ and $B = \frac{0.7}{3} + \frac{0.3}{5}$.

Verify De -Morgan's Law $\overline{A \cup B} = \overline{A} \cap \overline{B}$.

Ans. Given: $A = \frac{0.8}{3} + \frac{0.9}{4} + \frac{0.6}{5}$ and

$$B = \frac{0.7}{3} + \frac{0.3}{5}$$

$$\therefore \quad A \cup B = \frac{0.8}{3} + \frac{0.9}{4} + \frac{0.6}{5}$$

$$\Rightarrow \overline{A \cup B} = \frac{0.2}{3} + \frac{0.1}{4} + \frac{0.4}{5}$$

Also $\overline{A} = \frac{0.2}{3} + \frac{0.1}{4} + \frac{0.4}{5}$ and

$$\overline{B} = \frac{0.3}{3} + \frac{0.7}{5}$$

$$\Rightarrow \overline{A} \cap \overline{B} = \frac{0.2}{3} + \frac{0.1}{4} + \frac{0.4}{5}$$

$$\therefore \quad \boxed{\overline{A \cup B} = \overline{A} \cap \overline{B}}$$ **Hence verified.**

Ex.5. **Let age in years = X = {5, 10, 20, 30, 40, 50, 60, 70}**

Let A, B, C be the fuzzy sets defined by

Infant = A = {0, 0, 0, 0, 0, 0, 0, 0}

Young = B = {1, 1, 0.8, 0.5, 0.2, 0.1, 0, 0}

Old = C = {0, 0, 0.1, 0.2, 0.4, 0.6, 0.8, 1}

Find: (i) $A \cup B$ (ii) $A \cap B$ (iii) $A \cup (B \cap C)$

(iv) $B \cup C$ (v) $B \cap C$

Soln. Here X = {5, 10, 20, 30, 40, 50, 60, 70}

$$A = \left\{\frac{0}{5}, \frac{0}{10}, \frac{0}{20}, \frac{0}{30}, \frac{0}{40}, \frac{0}{50}, \frac{0}{60}, \frac{0}{70}\right\}$$

$$B = \left\{\frac{1}{5}, \frac{1}{10}, \frac{0.8}{20}, \frac{0.5}{30}, \frac{0.2}{40}, \frac{0.1}{50}, \frac{0}{60}, \frac{0}{70}\right\}$$

$$C = \left\{\frac{0}{5}, \frac{0}{10}, \frac{0.1}{20}, \frac{0.2}{30}, \frac{0.4}{40}, \frac{0.6}{50}, \frac{0.8}{60}, \frac{1}{70}\right\}$$

(i) $A \cup B = \max\{\mu_A(x), \mu_B(x)\}$

$$= \left\{\frac{1}{5}, \frac{1}{10}, \frac{0.8}{20}, \frac{0.5}{30}, \frac{0.2}{40}, \frac{0.1}{50}, \frac{0}{60}, \frac{0}{70}\right\}$$

(ii) $A \cap B = \min \{\mu_A(x), \mu_B(x)\}$

$$= \left\{\frac{0}{5}, \frac{0}{10}, \frac{0}{20}, \frac{0}{30}, \frac{0}{40}, \frac{0}{50}, \frac{0}{60}, \frac{0}{70}\right\}$$

(iii) $B \cap C = \min \{\mu_B(x), \mu_C(x)\}$

$$= \left\{\frac{0}{5}, \frac{0}{10}, \frac{0.1}{20}, \frac{0.2}{30}, \frac{0.2}{40}, \frac{0.1}{50}, \frac{0}{60}, \frac{0}{70}\right\}$$

$$\therefore \quad A \cup (B \cap C) = \left\{\frac{0}{5}, \frac{0}{10}, \frac{0.1}{20}, \frac{0.2}{30}, \frac{0.2}{40}, \frac{0.1}{50}, \frac{0}{60}, \frac{0}{70}\right\}$$

(iv) $$B \cup C = \left\{\frac{1}{5}, \frac{1}{10}, \frac{0.8}{20}, \frac{0.5}{30}, \frac{0.4}{40}, \frac{0.6}{50}, \frac{0.8}{60}, \frac{1}{70}\right\}$$

(v) $$B \cap C = \left\{\frac{0}{5}, \frac{0}{10}, \frac{0.1}{20}, \frac{0.2}{30}, \frac{0.2}{40}, \frac{0.1}{50}, \frac{0}{60}, \frac{0}{70}\right\}$$ **..Ans.**

Q.7. If $A = \left\{\frac{1}{2}, \frac{0.5}{3}, \frac{0.3}{4}, \frac{0.2}{5}\right\}$ and $B = \left\{\frac{0.5}{2}, \frac{0.7}{3}, \frac{0.2}{4}, \frac{0.4}{5}\right\}$ are fuzzy sets on universe $X = \{1, 2, 3, 4, 5\}$, then find $\overline{A}$, $\overline{B}$, $A \cup \overline{A}$ and $B \cap \overline{B}$.

Ans. Given:

X	A	B
1	0	0
2	1	0.5
3	0.5	0.7
4	0.3	0.2
5	0.2	0.4

$\therefore \overline{A} = \left\{\frac{1}{1}, \frac{0}{2}, \frac{0.5}{3}, \frac{0.7}{4}, \frac{0.8}{5}\right\}$ and $\overline{B} = \left\{\frac{1}{1}, \frac{0.5}{2}, \frac{0.3}{3}, \frac{0.8}{4}, \frac{0.6}{5}\right\}$

$\Rightarrow A \cup \overline{A} = \frac{1}{1} + \frac{1}{2} + \frac{0.5}{3} + \frac{0.7}{4} + \frac{0.8}{5} = 2$ and

$B \cap \overline{B} = \frac{0}{1} + \frac{0.5}{2} + \frac{0.3}{3} + \frac{0.2}{4} + \frac{0.4}{5} = 0.48$ ....Ans.

Ex.15. Find $A \cap B$, $A \cup B$, if

$$A = \frac{0.9}{1} + \frac{0.7}{3} + \frac{0.2}{4} + \frac{0.3}{6} \text{ and } B = \frac{0.1}{2} + \frac{0.4}{3} + \frac{0.5}{4} + \frac{0.8}{5}$$

Soln. Given: $A = \frac{0.9}{1} + \frac{0.7}{3} + \frac{0.2}{4} + \frac{0.3}{6}$ and

$$B = \frac{0.1}{2} + \frac{0.4}{3} + \frac{0.5}{4} + \frac{0.8}{5}$$

$\therefore \quad A \cap B = \min \{\mu_A(x), \mu_B(x)\}$

$$= \frac{0.9}{1} + \frac{0.1}{2} + \frac{0.4}{3} + \frac{0.2}{4} + \frac{0.8}{5} + \frac{0.3}{6} \text{ and}$$

$A \cup B = \max \{\mu_A(x), \mu_B(x)\}$

$$= \frac{0.9}{1} + \frac{0.1}{2} + \frac{0.7}{3} + \frac{0.5}{4} + \frac{0.8}{5} + \frac{0.3}{6} \text{ ....Ans.}$$

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Ex.16. Let U = {a, b, c, d} be the domain and A and B be fuzzy sets on U as given below:

	a	b	c	d
A	0.5	0.8	0.0	0.3
B	0.2	1.0	0.1	0.7

Find:

(i) $A \cup B$ (ii) $A \cap B$ (iii) A' (iv) B'.

Also prove that (v) $(A \cup B)' = A' \cap B'$.

Soln. Given:

	a	b	c	d
A	0.5	0.8	0.0	0.3
B	0.2	1.0	0.1	0.7

(i) $A \cup B(a) = \max [A(a), B(a)]$

$= \max (0.5, 0.2) = 0.5$

Similarly, $A \cup B(b) = 1.0$,

$A \cup B(c) = 0.1$

and $A \cup B(d) = 0.7$

Thus finally we obtain

	a	b	c	d
$A \cup B$	0.5	1.0	0.1	0.7

Note that this table could have been arrived at from the first table by taking the maximum of each column separately. In a similar way, performing the minimum of each column separately, we obtain $A \cap B$ as follows :

(ii)

	a	b	c	d
$A \cap B$	0.2	0.8	0.0	0.3

(iii) Subtracting each entry of A from 1 in the first table, we obtain A'

	a	b	c	d
A'	0.5	0.2	1.0	0.7

(iv) Similarly,

	a	b	c	d
B'	0.8	0	0.9	0.3

(v) Subtracting each entry of $A \cup B$ from 1, we obtain $(A \cup B)'$ as

	a	b	c	d
$(A \cup B)'$	0.5	0.0	0.9	0.3

Also taking minimum of each colum in (iii) & (iv) above we obtain $A' \cap B'$ as

	a	b	c	d
$A' \cap B'$	0.5	0.0	0.9	0.3

Thus $(A \cup B)' = A' \cap B'$ ..... **Ans.**

Ex.17. Let $A = \frac{0.5}{3} + \frac{1}{5} + \frac{0.6}{7}$, $B = \frac{0.7}{3} + \frac{0.6}{5}$ **be two fuzzy sets.**

Find: (i) A + B (ii) AB (iii) $A \oplus B$ (iv) $A \ominus B$

Soln. Given: $A = \frac{0.5}{3} + \frac{1}{5} + \frac{0.6}{7}$ and $B = \frac{0.7}{3} + \frac{0.6}{5} + \frac{0}{7}$

(i) $\mu_{A+B}(x) = [\mu_A(x) + \mu_B(x)] - [\mu_A(x) \cdot \mu_B(x)]$

$\therefore \quad A + B = \frac{0.85}{3} + \frac{1}{5} + \frac{0.6}{7}$

(ii) $\mu_{A\cdot B}(x) = \mu_A(x) \cdot \mu_B(x)$

$$\therefore \quad A \cdot B = \frac{0.35}{3} + \frac{1}{5} + \frac{0}{7}$$

(iii) $\mu_{A \oplus B} = \min [1, (\mu_A(x) + \mu_B(x))]$

$$\therefore \quad A \oplus B = \frac{1}{3} + \frac{1}{5} + \frac{0.6}{7}$$

(iv) $\mu_{A \ominus B}(x)$ $A = \left\{\frac{0.2}{x_1}, \frac{0.3}{x_2}, \frac{0.5}{x_3}\right\}$ and $B = \left\{\frac{0.5}{y_1}, \frac{0.6}{y_2}, \frac{0.7}{y_3}\right\}$

$$\therefore \quad A \ominus B = \frac{0}{3} + \frac{0.4}{5} + \frac{0.6}{7} \text{ ....Ans.}$$

Ex.18. **For the fuzzy sets** $A = \frac{0.2}{x_1} + \frac{0.5}{x_2} + \frac{0.6}{x_3}$ **and**

$B = \frac{0.1}{x_1} + \frac{0.4}{x_2} + \frac{0.5}{x_3}$, **find: (i)** $A \oplus B$ **(ii)** $A \ominus B$

Soln. **Given:** $A = \frac{0.2}{x_1} + \frac{0.5}{x_2} + \frac{0.6}{x_3}$ and $B = \frac{0.1}{x_1} + \frac{0.4}{x_2} + \frac{0.5}{x_3}$

(i) $\mu_{A \oplus B}(x) = \min [1, (\mu_A(x) + \mu_B(x))]$

$$\therefore \quad A \oplus B = \frac{0.3}{x_1} + \frac{0.9}{x_2} + \frac{1}{x_3}$$

(ii) $\mu_{A \ominus B}(x) = \max [0, (\mu_A(x) - \mu_B(x))]$

$$\therefore \quad A \ominus B = \frac{0.1}{x_1} + \frac{0.1}{x_2} + \frac{0.1}{x_3} \text{ ....Ans.}$$